

How Should I React? Learning to Face Chemical Risks with Virtual Reality

Pierre Raimbaud¹
Guillaume Lavoué¹
Bertrand Massot³

Matthieu Blanchard¹
Janine Jongbloed²
Jean-Pierre Cloarec³
Audrey Serna¹

Eliott Zimmermann¹
Rawad Chaker²
Jean-Louis Leclercq³
Élise Lavoué¹

Sophie Villenave¹
Matthieu Mesnage³
Valentin Midez¹

¹ Ecole Centrale de Lyon, CNRS, INSA Lyon, UCBL, Université Jean Moulin Lyon 3, iaelyon school of Management, LIRIS, UMR5205, Lyon, France *

² Laboratoire ECP, Université Lumière Lyon 2, 86 rue Pasteur, F-69365, Lyon, France

³ INSA Lyon, Ecole Centrale Lyon, Université Claude Bernard Lyon 1, CPE Lyon, CNRS, INL UMR 5270, 69100, Villeurbanne, France



Figure 1: Virtual Reality (VR) chemical risk training: A) Pouring acid and extinguishing fire in VR; B) Thermal and smell devices for multisensory stimuli; C) Sensors integrated to controllers to capture physiological signals; D) Reflexive feedback for learners in VR.

ABSTRACT

Our project proposes to use Virtual Reality (VR) to train chemists to react to incidents. It explores the effects of multisensoriality on learner VR sense of presence and immersion, and the effects of visualising (in/out of VR) behavioural and physiological indicators on learner reflexivity. Multisensoriality implementation, robust capture of behavioural and physiological data, and the design of relevant indicators are our main challenges. Three situations involving chemical incidents have been developed, engaging the learner to react accordingly. We present the first implementations of this VR training environment and the results of our first user studies.

Index Terms: Virtual Reality, Training, User Experience, Multisensory Experience, Behaviours, Physiological Signals, Reflexivity

1 INTRODUCTION

Virtual Reality (VR) technology is used in more and more fields today, and the literature has notably shown its benefits for training in diverse contexts [6]. Wu et al. [5] performed a meta-analysis of training studies in VR, to show its effectiveness in improving the learning performance of its users, compared to traditional methods (e.g., articles, presentations, videos). Moreover, another noticeable interest of using VR for training is its ability to expose the users to plausible hazardous situations, without really exposing them to these situations. This, coupled with the ecological validity of VR i.e., the fact that users reproduce real behaviours in VR, can explain its popularity for safety training [4]. However, open questions remain, notably about the sensory modalities to transmit plausible and effective stimuli to VR users and how they can affect their experience, as well as the ways to provide feedback to users on their experience, and how it could be beneficial for learning. In this frame, we present here our project that focuses on a multisensory reflexive Virtual Environment (VE) to train chemists to react to risks.

2 RESEARCH QUESTIONS AND CHALLENGES

This project proposes to study two main exploratory research questions: i) understanding to what extent multisensory stimuli improve learner presence and immersion, through the enhancement of the

embodied experience; ii) understanding how the visualisation of behavioural and physiological indicators during and after the VR experience can improve learner reflexivity. Exploring these questions induces the following challenges: i) the integration and evaluation of multisensory stimuli in VR, and their interplay; ii) the development and evaluation of sensors that are non-intrusive, real-time and robust to 6-Degrees-of-Freedom VR, enabling the capture of physiological data that can be used to support reflexive processes; iii) the analysis and visualisation of learners' behaviours, both in VR (immediate feedback) and in debriefing environments.

3 FIRST IMPLEMENTATIONS

3.1 VR chemical scenarios: scriptwriting, development

Three main chemical situations have been designed for our project, by chemists and researchers in education, since our VR project targets real chemist training usages such as at universities. The first chemical situation (1) consists in preparing dilutions of sulphuric acid at different concentrations; the second one (2) in titrations of unknown pH solutions of acetic acid; the third one (3) in dilutions of a 90° alcohol to prepare 70° bottles. They have been selected as they are common and easily enable typical chemical risks. In (1), a too fast dilution would provoke bubbling, with dangerous splashes and steams; in (2), a bad ventilation by the fume hood would provoke the inhalation of unsafe steams; (3) is not particularly risky, but will be perturbed by a fire due to previous manipulations.

A scriptwriting has been designed to fully describe the expected reactions depending on the incident provoked and the virtual personal protective equipment worn by the learner, as well as the expected consequences of not reacting. This has contributed to guide the development of the VR application on Unity game engine, with a system of steps and conditions validation, feedback, and replacement to safe or hazardous situations to train to react to risks.

3.2 Multisensory VR: smell and thermal devices

First, we reproduced a smell machine designed by Javerliat et al. [1] that is wearable under a VR Head-Mounted Display and capable of diffusing odours at a desired intensity. With this, we enable and control smell diffusion from the VR application. In the frame of this project, we improve this machine to enable the diffusion of two different odours, to allow the diffusion of vinegar-like smell to alert the learner about the virtual emission of acetic acid steam for Situation 2, and fire-like smell for Situation 3.

We also designed a thermal device based on infrared lamps to simulate a fire. This one is controllable by the VR application,

*Corresponding author e-mail: pierre.raimbaud@liris.cnrs.fr

3.3 User physiological and behavioural data collection

To support reflexive processes, user data needs to be captured from the VR experience, to be visualised in or out of the VE. For that, an open-source and open-hardware ecosystem for physiological sensing has been developed. Our toolkit focuses on reliable monitoring of cardiac activity via electrocardiogram (ECG) and electrodermal activity (EDA) through skin conductivity, with the aim of studying the autonomic nervous system's responses in VEs. This system enables some objective, quantitative analyses of emotional responses, including stress, mental workload, and habituation, during the scenarios and learning tasks created for the project. To minimise sensor intrusiveness and avoid the need for multiple systems to be worn by the user, the sensors have been designed for seamless integration into VR controllers. This allows for signal collection directly as the learner holds them. Both sensors feature Bluetooth Low Energy communication, ensuring freedom of movement, while a Python package on the computer facilitates the connection to the sensors, the signal collection and storage, and the data transmission to the VR application for real-time biofeedback. We also developed a software toolbox to enable the automatic and standardised collection of all behavioural data from user actions in the VE, for all our studies. The toolbox ensures behavioural data synchronisation with the physiological ones, notably by providing an easy integration of the sensors signals through the Lab Streaming Layer framework.

4 PRELIMINARY STUDIES

4.1 Thermal stimuli effect on user quality of experience

The current literature on thermal stimuli in VR has not focused on evaluating the effect of dynamic changes of heat according to user action on the presence and haptic experience. As this aspect is paramount to design and implement our chemical training VE, we conducted a user experiment. In this study, all our users were exposed to three conditions: a control one without thermal feedback, a static thermal one, and a dynamic thermal one (changes due to user actions). Questionnaires on general presence, spatial location, sense of agency and haptic experience were used here. Our results show that, despite a high sense of presence in the control condition, adding thermal feedback still increased the presence. No differences were observed between static and dynamic conditions, which could be explained by strong interpersonal differences in thermal sensitivity, which correlates with the haptic experience we measured.

4.2 Physiological sensors: signal quality and comfort

Currently, there are no recommendations in the literature to ensure signal quality when capturing physiological data during VR immersions with freedom of movement [2]. Moreover, the current standard for capturing EDA is on the fingers, whereas the most common interactions in VR rely on hand controllers. To contribute to address this issue, we ran a user experiment where they were asked to perform series of five different gestures (one being motionless as control) while being equipped with EDA and ECG sensors. Three positions were tested for EDA (palm, wrist, fingers), two for ECG (left and right side of chest), and each series was conducted with different electrode position combinations, by all participants. We analysed both the signal quality depending on the gestures and the sensors positions. We also asked the users about the level of intrusiveness and discomfort of the sensors. We recruited 36 participants (13 self-reported as female, 23 male, average age 33.3 ± 11.5).

Our results show that the palm position is a good alternative to the finger one, with equivalent or significantly better quality for all the different gestures, despite a systematic drop in quality when opening or closing the hand for all sensor positions. For ECG signals, significant difference was only observed between the motionless movement and the other gestures. Moreover, the left-side chest position always gave the best signal quality. Regarding the comfort, the palm and wrist positions for EDA were rated as significantly more comfortable and less intrusive than the finger one. No significant differences were observed for ECG sensor positions.

4.3 Immediate feedback effect on learner's presence and performances in VR

The aim of immediate feedback is to get users to reflect on the action in progress so they readjust and modify their actions accordingly [3]. The main challenge is to encourage reflection while preserving a high quality of experience, notably the sense of presence.

As a preliminary study, we ran a user experiment with immediate feedback given through stars displayed in VR to reward correct responses during a chemical manipulation, such as accurately identifying the pH of a solution. We also used a progress bar that incrementally increased for each correctly performed action. This visual feedback was complemented by auditory signals to draw users' attention to the appearance of a star or an increase in the progress share. Users completed the IPQ questionnaire after the VR experience to evaluate their feeling of presence and immersion. Alongside, we developed an ad-hoc behavioural sensor system in Unity to track high-level interaction data, such as handled objects, scored points, and taken actions. We recruited 46 participants (17 self-reported as female, 29 male, average age 23.4 ± 6.9), with half receiving feedback (F group), and half without (WF group).

Our results showed that the sense of spatial presence was significantly higher for the WF group when performing scenario tasks. No significant differences were observed on user experience involvement and realism measures. For performance, measured counting correctly performed tasks, we observed that the F group obtained a significantly lower number of correct answers. We ran deeper analyses on the evolution of correct answers between similar subtasks (i.e., with the same objective) and observed a significant increase in the number of correct answers between the first and second subtasks for the F group – no significant evolution was observed for the WF group. In conclusion, our findings highlight that the integration of immediate feedback reduces the sense of presence as expected, but that this feedback allows the user to improve on repetitive tasks.

5 CONCLUSION AND FUTURE WORK

We developed a multisensory VE for chemical risk training. Our study on thermal feedback showed its effect on presence. Our thermal device could be improved by using local actuators to simulate local heat. We found in our other study that EDA sensors can be put in users' palm for signal quality and comfort, but we still need to improve the quality when closing hands. Finally, to complete our study on reflexivity with behavioural data, we will work on integrating physiological signals during and after VR experience.

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REFERENCES

- [1] C. Javerliat, P.-P. Elst, A.-L. Saive, P. Baert, and G. Lavoué. Nebula: An affordable open-source and autonomous olfactory display for vr headsets. In *Proceedings of the 28th ACM symposium on virtual reality software and technology*, pp. 1–8, 2022. 1
- [2] P. Lopes, N. Fachada, M. Fonseca, H. Gamboa, and C. Quaresma. Modelling physiological sensor noise to movement-based virtual reality activities. In *BIOSTEC (1)*, pp. 778–785, 2024. 2
- [3] L. Mandouit and J. Hattie. Revisiting “the power of feedback” from the perspective of the learner. *Learning and Instruction*, 84:101718, 2023. 2
- [4] D. Scorgie, Z. Feng, D. Paes, F. Parisi, T. Yiu, and R. Lovreglio. Virtual reality for safety training: A systematic literature review and meta-analysis. *Safety science*, 171:106372, 2024. 1
- [5] B. Wu, X. Yu, and X. Gu. Effectiveness of immersive virtual reality using head-mounted displays on learning performance: A meta-analysis. *British journal of educational technology*, 51(6):1991–2005, 2020. 1
- [6] B. Xie, H. Liu, R. Alghofaili, Y. Zhang, Y. Jiang, F. D. Lobo, C. Li, W. Li, H. Huang, M. Akdere, et al. A review on virtual reality skill training applications. *Frontiers in Virtual Reality*, 2:645153, 2021. 1